

Tomographic Reconstruction of the Neutron Time–Energy Spectrum From a Dense Plasma Focus

Jared Catenacci, Derek Constantino, Brady Gall, and Aaron Luttman[✉]

Abstract—A dense plasma focus (DPF) nuclear fusion device is an attractive pulsed neutron source in many applications, due to its relatively large neutron yield, produced in a short time duration. To design a DPF that generates neutrons within a specified time profile, or that generates a neutron energy spectrum with specific properties, it is necessary to characterize and model the results of the fusion process releasing the neutrons. The time–energy spectrum of fusion neutrons is an ideal quantity used in validating multiphysics codes that simulate the pinch and fusion processes, because it is a quantity that requires physics of each of the stages leading up to and ending in the fusion reaction to be simulated correctly. In particular, since DPF fusion neutrons are not monoenergetic—and there can often be several fusion pinches creating neutrons—a computer simulation matching high quality neutron spectrum measurements provides confidence in the fidelity of the simulation. In order to make such a comparison, it is first necessary to have quality measurements from which to infer the spectrum. In this work, we project neutron spectroscopy as the classical tomographic inverse problem from neutron time-of-flight data at multiple distances, but enhanced by using an additional measurement of the time profile of the fusion pinch near its source and by using detector pairs set up in a geometry that allows for scatter background subtraction. The detector pairs enhance the quality of the time-of-flight measurements, and the additional constraint posed by the measured time profile allows for reconstructions discretized as finely as the time measurements and in energy as finely as 100 keV, without the problem being underdetermined. We present results from a deuterium-fueled DPF at the U.S. Department of Energy’s Nevada National Security Site and show that we can infer the time–energy spectrum from our measurements for both single and multi-pinch fusion reactions with equal fidelity.

Index Terms—Dense plasma focus (DPF), time-of-flight (ToF) spectroscopy.

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I. INTRODUCTION

A DENSE plasma focus (DPF) is a machine that produces a plasma capable of achieving nuclear fusion, resulting in the emission of X-rays and neutrons [1], [2]. As a pulsed X-ray or neutron source, the DPF has a wide range of applications, including active interrogation of materials and neutron radiography (imaging). In many cases such measurements require the DPF-generated neutron pulse to have a specific time profile and to meet a minimum yield threshold. Essentially, the applications demand a copious amount of neutrons, especially in a short time window. In order to design a DPF system that can repeatably obtain pulses with the desired characteristics, it is crucial to have predictive models with associated computationally tractable simulation capabilities that are validated against reliable measurements of the fusion reaction properties. Among the most important quantities to understand is the neutron time–energy spectrum, as it is a single quantity that is a direct result of the entire multiphysics of the DPF process and is notoriously difficult to simulate [3]–[7] with high fidelity.

The most common approach to fusion neutron spectroscopy is to take time-of-flight (ToF) measurements at multiple detectors, each placed at different distances from the pinch, and to invert a tomographic model—essentially the Radon transform—to reconstruct the energy density from the measurements [8]. There are well-established computational methods in the literature for inverting the Radon transform in this context [9]–[11], but even the most popular [12], [13] suffer from poor time and energy resolution when there are few detectors or when the noise level is high.

In this work, we pose a novel approach for reconstructing the time-dependent neutron energy spectrum that directly incorporates a measurement of the radiation pulse near the source, along with modified ToF detector pairs designed to reduce the neutron scatter background. The detector pairs substantially improve the quality of the ToF data, and the near-source measurement constrains the spectrum reconstruction significantly more than the ToF measurements alone. The addition of the close-range neutron data provides the ability to reconstruct the time–energy spectrum with sub-nanosecond in time and estimated sub-100 keV in energy discretizations. Our reconstruction method is applied to data collected at the U.S. Department of Energy’s Nevada National Security Site (NNSS) DPF Laboratory in North Las Vegas, NV. We are able to reconstruct the neutron creation

time and neutron creation energy spectrum—observing the energy shifts and broadening at around 2.45 MeV that are common to beam–target interactions [14]—and observe how the energy density changes temporally, including validation of our calculations of peak neutron energy using standard ToF principles.

In Section II, we develop the mathematical model and define the reconstruction method that incorporates the neutron pulse profile at close range to constrain the inverse problem. Section III gives the details of the modified ToF detector systems and the additional data processing and calibration methods applied. In Section IV, we present the time–energy reconstructions and discuss the results, with a conclusion in Section V.

II. MODEL FORMULATION

Neutron tomography is a method for analyzing neutron creation events, given data from detectors that measure neutron flux over time, placed at different distances from the source. A single neutron detector is generally incapable of determining the energy of an absorbed particle, since neutrons arriving directly at a detector at the same time only have the same energy if they were also created at the same time. Detectors at multiple distances can overcome this, and the recovery of neutron kinetic energy distributions from ToF data has been an active area of research for several decades [8].

If t is the time at which a particle is emitted, t' is the time at which the particle is detected, X is the distance from the source to the detector, and E is the kinetic energy of the particle, then the goal is to reconstruct the time-dependent energy distribution, $f(t, E)$, from measurements, $S(X, t')$. The basic mathematical model for ToF-based spectroscopy, introduced in [10], is given by

$$S(X, t') = \int_{-\infty}^{\infty} \int_0^{\infty} f(t, E) \delta\left(t - t' + \frac{X}{\sqrt{2E/m_0}}\right) dE dt \quad (1)$$

where δ is the Dirac delta and m_0 is the particle's rest mass. It is common to assume that the time-dependent energy distribution $f(t, E)$ is fully separable, so that $f(t, E) = f_t(t) f_E(E)$, and, additionally, assumptions are often made imposing a parametric form on both $f_t(t)$ and $f_E(E)$. These assumptions greatly simplify the computational aspects of the reconstruction, with the tradeoff being that the energy density is fixed over time (up to a time-dependent scale) and one must assume that the chosen parametric form of the distributions is approximately correct. This can be especially problematic in cases of multiple pinches, since the assumption of separability imposes a solution whose neutron energy density is the same for all pinches. In order to allow for the shape of the neutron energy distribution to change over time, an alternate and popular approach is to drop the assumption of a parametric form for $f_t(t)$ and $f_E(E)$, fully discretize in both time and energy, and reconstruct using a Monte Carlo method [9], [12], [13]. The challenge of this approach is that many detectors are required in order to constrain the problem with sufficiently fine discretization in t and E .

To circumvent these issues, we take a modified approach that is a hybrid between classical and Monte Carlo reconstruction methods. We assume that the time-dependent energy

distribution is partially separable, that is, $f(t, E) = g(t) h(E(t))$, where $g(t)$ is taken from a direct measurement of the neutron pulse time profile at close range to the pinch, and the energy distribution $h(E(t))$ is constant in time over fixed subintervals of the fusion reaction being characterized. This allows us to avoid placing parametric assumptions on the distributional form and to reconstruct more finely in time and energy, and it also allows for variation in the energy density over intervals of time relevant to the dynamics of neutron generation, i.e., multiple pinches.

Corresponding to the time sampling of the digitizers, we discretize in time at points $\{t_j\}_{j=1}^J$, and the first step in the model is to select a subset of those points $\{r_0 = t_1, r_1, r_2, \dots, r_{K-1}, r_K = t_J\}$, giving intervals $\tau_k = [r_{k-1}, r_k)$ for $k = 1, \dots, K$, that encompass the different stages of the fusion reaction. For example, if there are two pinches, then typically K is chosen to be equal to 3, so the first time interval encompasses the first pinch; the second time interval encompasses the second pinch; and the third time interval encompasses the tail of the reaction. The neutron energy distribution is then treated as separable on each time interval.

Second, energy is discretized by selecting points $\{E_1, \dots, E_M\}$. In the work that follows, $\{E_m\}_{m=1}^M$ is a uniform discretization from 1.45 to 3.45 MeV, with M typically ranging from 25 to 30.

The proposed model for the time–energy distribution is then given by

$$f(t, E) = g(t) \sum_{k=1}^K h_k(E) \mathbb{1}_{\tau_k}(t) \quad (2)$$

where $g(t)$ is the measured probability density function for the neutron creation time and $\mathbb{1}_{\tau_k}$ is the indicator function over the interval τ_k .

The function $h_k(E)$ is the energy distribution over the interval τ_k , and to allow for a generic representation of $h_k(E)$, we choose a nonparametric approximate scheme comprising a linear combination of Dirac masses

$$h_k(E) \approx \sum_{m=1}^M w_{km} \delta(E - E_m) \quad (3)$$

where w_{km} is the weighting coefficient. Substituting into (2) yields

$$f(t, E) \approx g(t) h(E(t)) = g(t) \sum_{k=1}^K \sum_{m=1}^M w_{km} \delta(E - E_m) \mathbb{1}_{\tau_k}(t) \quad (4)$$

and incorporating the time and energy distributions into (1) gives

$$\begin{aligned} S(X, t') &= \int_0^{\infty} g\left(t' - \frac{X}{\sqrt{2E/m_0}}\right) \\ &\quad \times \sum_{k=1}^K \sum_{m=1}^M w_{km} \delta(E - E_m) \mathbb{1}_{\tau_k}\left(t' - \frac{X}{\sqrt{2E/m_0}}\right) dE \\ &= \sum_{k=1}^K \sum_{m=1}^M w_{km} g\left(t' - \frac{X}{\sqrt{2E_m/m_0}}\right) \mathbb{1}_{\tau_k}\left(t' - \frac{X}{\sqrt{2E_m/m_0}}\right). \end{aligned} \quad (5)$$

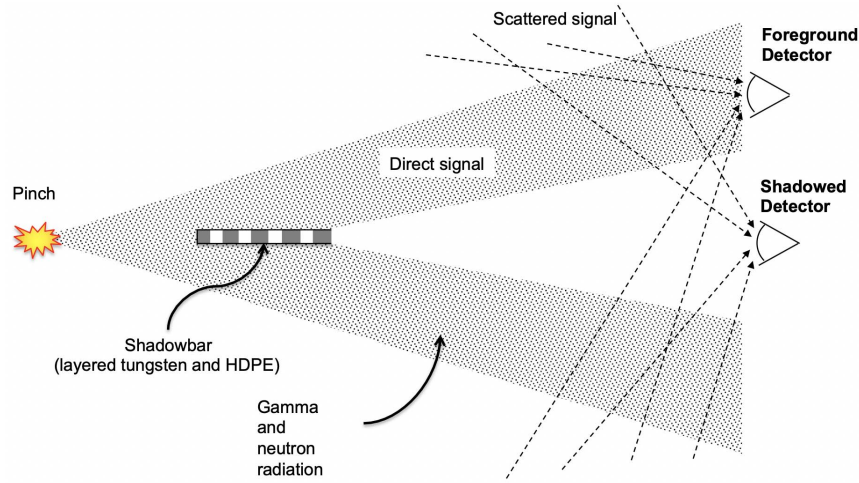


Fig. 1. Shadow bar system configuration. The two detectors are 60 cm apart for the pair at 10 m, 27 cm for the pair at 14 m, 31 cm for the pair at 18 m, and 39 cm apart for the pair at 22 m.

This model cannot be compared directly to a measured signal, as the distance of the detector from the source, X , the total number of particles created, N , the detector efficiency α , and the detector surface area orthogonal to the line of sight, A must be accounted for. Incorporating these effects, and assuming L detectors at distances $\{X_1, \dots, X_L\}$, gives the detector model

$$S_\ell^*(X_\ell, t') = \frac{NA\alpha_\ell}{4\pi X_\ell^2} S(X_\ell, t') \quad (6)$$

for $\ell = 1, \dots, L$.

Given the close-range data $\{g(t_j)\}_{j=1}^J$ and the ToF detector data $\{y(X_\ell, t_j)\}_{j=1, \ell=1}^{J, L}$, we wish to solve

$$\hat{q} = \underset{q \in \mathcal{Q}}{\operatorname{argmin}} J(q) := \sum_{\ell=1}^L \sum_{j=1}^J (y(X_\ell, t_j) - S_\ell^*(X_\ell, t_j; q))^2. \quad (7)$$

The vector q contains the unknown parameters, $\{w_{km}\}_{k=1, m=1}^{K, M}$ and $\{\alpha_\ell\}_{\ell=1}^L$, and we optimize over the set of admissible parameters, $\mathcal{Q} \subset \mathbb{R}_+^p$, where $\mathbb{R}_+$ represents the non-negative real line, and $p = KM + L$ is the total number of parameters to be estimated. Note that there are $(L+1)J$ data points, from the L ToF detectors plus the close-range detector, and we choose K and M so that $KM + L \ll (L+1)J$. For the data shown below, $J \sim 10^6$, $K = 3$, $L = 4$, and $M = 30$, so the inequality is satisfied by orders of magnitude.

III. EXPERIMENTAL SETUP

The NNSS operates a deuterium deuterium (D-D) DPF at its North Las Vegas facility; see [5] for details on the DPF system. In the sections that follow we provide the details of the ToF shadow bar systems and the close-range detector used to reconstruct the neutron time-energy spectrum.

A. Shadow Bar System

Typically, the pulses generated by the DPF are recorded using individual ToF detectors that respond to incident photons and neutrons as a function of time. However, in many

situations the measured neutron/photon pulse is heavily contaminated with background scatter. To diminish the impact of scattered particles on the reconstruction, a so-called “shadow bar” system is employed. The shadow bar configuration has two ToF detectors fielded side-by-side; see Fig. 1. The foreground detector, denoted FG, is exposed to the direct beam of neutrons and photons and also to scattered neutrons and photons. The shadowed detector, denoted SB, is exposed only to the scattered signal, being shadowed from the direct beam by a cylindrical bar of tungsten and high density polyethylene (HDPE). The cylindrical bar was positioned approximately equidistant between the detectors and DPF pinch for each detector pair’s line-of-sight. This position was determined using a stochastic scattering model and validated empirically. Thus, subtracting the second signal from the first gives a measure of the direct neutron beam. (See the appendix for the detector specifications.)

Since the neutron beam, and, in particular, the scatter field are not entirely isotropic, the detectors must be placed in close proximity to each other and, ideally, away from major scatter sources. In principle, the signal measured from the shadowed detector can be subtracted from the signal measured by the foreground detector and what remains is only a signal measuring the direct neutron beam. However, due to differences in the detector systems, the signal from the shadowed detector must be scaled in order to perform the subtraction in a meaningful way. In order to determine the scaling, several calibration measurements are taken without the shadow bar in place, so both detectors measure the direct neutron beam and the scatter background. These calibration measurements result in signals sharing the same characteristics, but are not well matched in amplitude due to differences in the two detector systems (see Fig. 2). In order to account for the difference in amplitudes, we scale the signal measured by the SB detector by the factor ρ to match the signal measured by the FG detector, which is computed by solving

$$\hat{\rho} = \operatorname{mean}\{\underset{\rho \in \mathbb{R}}{\operatorname{argmin}} \|y_{\text{FG}} - \rho y_{\text{SB}}\|^2\} \quad (8)$$

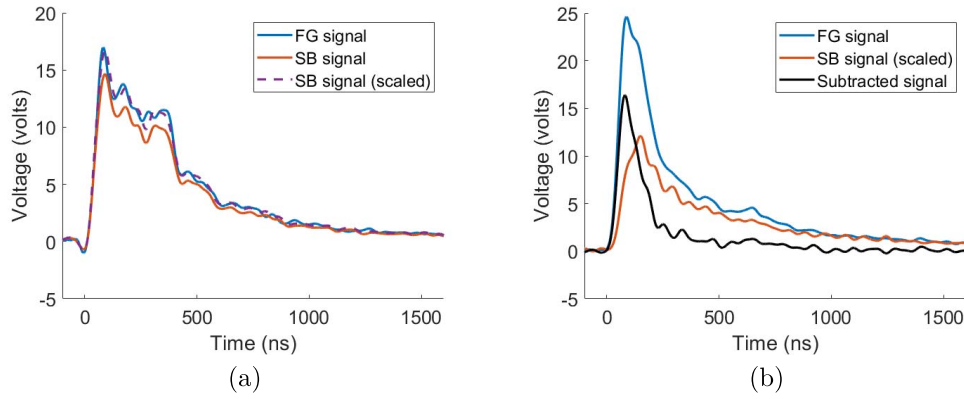


Fig. 2. (a) Example of a neutron pulse measured by the FG and SB detectors in a shadow bar system configured for calibration. The SB detector is unshielded by the shadow bar, thus both detectors have a direct line of sight to the pinch. The dashed purple line is the signal measured by the SB detector after scaling to match the FG detector according to (8). (b) Example of the neutron pulse measured by an FG and SB detector where the SB detector has the shadow bar in place. The SB signal was scaled according to the calibration factor, and the subtracted signal, representing the neutron pulse with the background removed, is shown as the solid black line.

TABLE I

AVERAGE OF THE ESTIMATED SCALING FOR EACH OF THE SB DETECTORS

Detector	Scale factor
SB - 10 m	1.13
SB - 14 m	0.84
SB - 18 m	0.98
SB - 22 m	1.07

where

- 1) y_{FG} and y_{SB} denote the FG and SB signals over the span of the neutron pulse, with the detector impulse responses deconvolved out,
- 2) $\|\cdot\|$ denotes the Euclidean norm, and
- 3) the mean is computed over the collection of calibration shots (typically around four or five shots).

Note that the scale factor must be computed for each shadow bar pair, and Table I shows the final scale factors for each of the four systems in our experimental setup. All the factors are fairly close to 1, which suggests that the pairs are well matched. Nonetheless, the scaling is essential prior to subtracting one signal from another.

An example of a calibration measurement and the resulting scaling of the SB signal are shown in Fig. 2(a). For the calibration, the shadow bar is not in place. Fig. 2(b) shows example SB (orange) and FG (blue) signals in a detector pair when the shadow bar is in place and the SB signal is scaled from the calibration shots. As expected, the FG signal is much larger than the SB signal, since the FG has both scatter and the direct neutron beam.

For the data used in this work, four shadow bar systems (detector pairs) were fielded. For ideal tomographic reconstruction, each of the system would share a single line of site to the fusion pinch, preferably in either the axial (collinear with the ion beam) or radial (orthogonal to the ion beam) direction. However, due to physical limitations of the facility where the DPF is housed, the shadow bar systems were placed between 63° and 76° , measured from the axial direction. The detectors are both neutron and photon sensitive and are lead-shielded in order to reduce the photon signal. The shielding does not

TABLE II

DISTANCE TO THE PINCH AND THE ANGLE MEASURED FROM THE AXIAL DIRECTION FOR EACH OF THE SHADOW BAR SYSTEM DETECTORS

Detector	Distance to pinch (m)	Angle
FG - 10 m	10.29	72.19°
SB - 10 m	10.29	76.06°
FG - 14 m	14.24	71.63°
SB - 14 m	14.29	70.56°
FG - 18 m	18.27	67.52°
SB - 18 m	18.30	66.52°
FG - 22 m	22.25	63.76°
SB - 22 m	22.27	64.77°

eliminate the photon signal, so the detector systems are placed at approximately 10, 14, 18, and 22 m away from the pinch, which are large enough distances that the gamma and neutron signals can be distinguished by ToF. The angles and placement of each of the eight detectors in the four systems are given in Table II. The positions are not optimal but were constrained by the facility. A diagram of the setup is shown in Fig. 3, showing the close-range detector, which is 25 cm from the DPF pinch, and the four shadow bar detector systems, which are placed on the other side of the DPF at the angles noted in Table II.

B. Close-Range System

In order to further constrain the spectrum reconstruction, we incorporate a close-range measurement to characterize the time profile of neutron creation. The measurement is taken by a single ToF detector approximately 25 cm from the fusion pinch. At this distance, the measured pulse has approximately the same shape as the neutron creation time profile since the pulse shape has not had enough distance to undergo ToF spreading. Thus, we have a direct measurement of the probability density function for the neutron creation time $g(t)$. This greatly reduces the computational complexity of the time-energy reconstruction and constrains the problem so that four shadow bar systems provide enough ToF data to reconstruct at a moderate energy resolution. The time

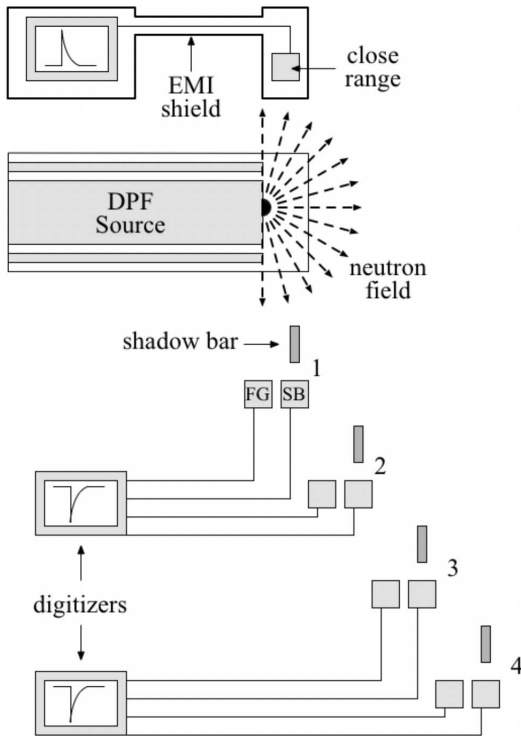


Fig. 3. Diagram of the experimental setup. The close-range detector is placed 25 cm from the pinch in the radial direction and the four shadow bar systems, comprising two neutron/photon detectors each, were placed at ranges of 10, 14, 18, and 22 m in the radial direction.

resolution is constrained by the sampling rate and bandwidth of the oscilloscope used to record the data, which are 2.5 GS/s and 1 GHz, respectively, as well as the detector impulse response, which is less than 2 ns. (See the appendix for the specifications of the close-range detector.)

IV. RESULTS AND DISCUSSION

In this section, we present the results of time-energy spectrum reconstructions from the NNSS North Las Vegas DPF for four independent shots as well as: 1) the impacts of the close-range measurement on the reconstructions, 2) the impact of the shadow bar systems, and 3) a comparison of the maximum neutron energy reconstruction to a straightforward, single-detector calculation of maximum neutron for each shot.

A. Impact of the Close-Range Measurement

It is well known that a DPF is capable of producing multiple neutron-generating pinches, and it is with some regularity that the NNSS DPF generates double pinch events. The second pinch is due to the formation of a second plasma sheath at the insulator, formed from residual gas left behind after the initial plasma sheath run-down [15]. This has been observed since the beginning of DPF research and is characterized as an occasional second peak in the neutron pulse time history [2].

Since we represent the time-dependent energy distribution with a nonparametric model, we are able to reconstruct the time-energy profiles for both single or double pinch events without modification to the model or algorithms. The top two graphs in Fig. 4, Trials 1 and 2, show reconstructions from

TABLE III
PEAK ENERGY IN THE RECONSTRUCTED TIME-ENERGY SPECTRUM AND TOTAL NEUTRON YIELD IN 4π FROM EACH TRIAL

Trial	Main Pinch (MeV)	Second Pinch (MeV)	Neutron Yield ($\times 10^{10}$)
1	2.42	—	3.72
2	2.47	—	4.28
3	2.41	2.59	4.38
4	2.48	2.48	4.60

single pinch events, and the bottom two show graphs, Trials 3 and 4 are reconstructions from double pinch events.

The peak (most probable) neutron energies of the four trials are given in Table III, which correspond to the brightest yellow points in each of the plots in Fig. 4 and are all near 2.45 MeV. This is expected, since the detector lines of sight are approximately radial to the pinch. Also noted are the total neutron yields in 4π for the trials.¹

Notice that the reconstructions for the double pinch events do not have identical energy densities for the first and second pinches, as the most probable energies for the two pinches are not necessarily the same, e.g., in Trial 3, where there is large difference between the peak energies for the first and second pinches. This is clearer in Fig. 5 where we compare the energy density from the first and second pinches for Trials 3 and 4. For Trial 3, the second pinch shows an increase in the density of higher energy neutrons around 2.7 MeV compared to the first pinch which has the highest concentration of neutrons near 2.45 MeV.

For Trial 4, the energy densities of the first and second pinches are similar at higher energies, with almost identical peak energies, but the lower energy portions of the spectrum are quite different. Resolving small differences in the energy density between the pinch events would not be possible using a standard approach to the tomographic problem, particularly since the pinches are separated by less than 50 ns from peak to peak. In fact, when the second pinch occurs so close to the first, the neutron pulse has a unimodal shape by the time it reaches the shadow bar systems due to ToF spreading. In Fig. 6, we show the close-range measurement for Trial 3 which is a clear double pinch, yet, the signal recorded by each of the shadow bar detectors gives no indication that the generated pulse has a double pinch feature, due to the ToF spreading. With only four shadow bar systems, reconstructing a temporal bimodal neutron pulse would be impossible without the inclusion of the additional constraints provided by the close-range measurement.

B. Impact of the Shadow Bar System

Without the shadow bar detectors to mitigate the effects of scatter, which are common in large experimental facilities and exist as permanent fixtures in the NNSS DPF Laboratory, our reconstructions would contain an artificially high density of lower energy neutrons (less than 2 MeV), as the direct

¹As the total neutron yields exceed 10^{10} neutrons in 4π for each trial, more than 10^4 neutrons are expected at the photomultiplier tube (PMT) at 22 m.

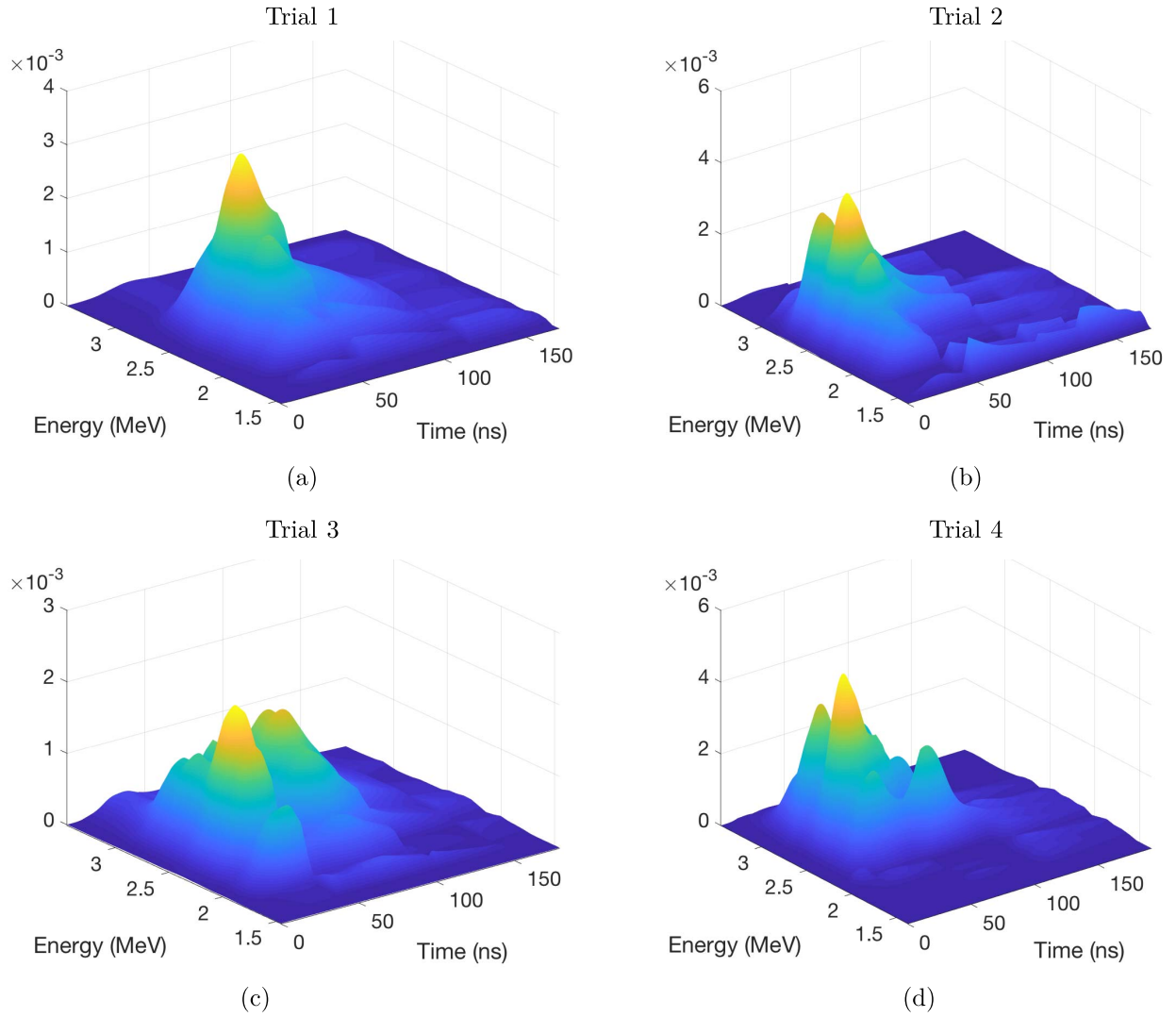


Fig. 4. Reconstructions of the time–energy spectrum using data from four different shots: (a) and (b) are reconstructions of single pinch events and (c) and (d) are reconstructions from double pinch events. The vertical axis represents the probability of a neutron being created at a given time with a given energy.

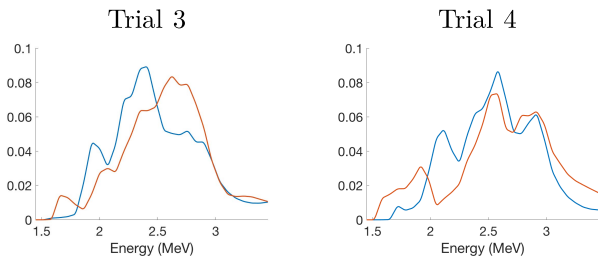


Fig. 5. For the trials with double pinches, the curves show the probability of a neutron being created with the given energy during the first (blue) or second (orange) pinches.

tomographic model associates lower energy to a scattered neutron's longer flight time, rather than associating a longer flight path to its longer time. To illustrate this, the reconstruction for Trial 1 is shown in Fig. 7, for the foreground-minus-background (top) and for the uncorrected signal of just the foreground detector (middle). Qualitatively the reconstructions are similar, but subtracting the top reconstruction from the middle reconstruction demonstrates clear differences, shown

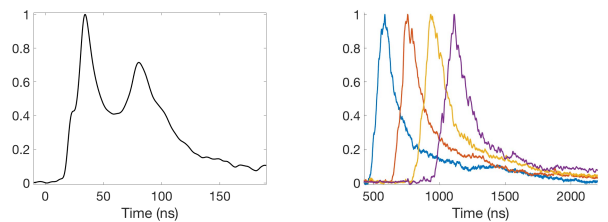


Fig. 6. Pinch profile as measured by the close-range detector (left) and measurements by the four shadow bar systems (right) for Trial 3. The double pinch structure observed by the close-range detector has been blurred to a single pulse by the time the neutrons reach the shadow bar detectors. The pulse is only 150 ns wide as measured by the close-range detector, but due to the ToF spreading the pulse is over 500 ns wide by the time it reaches the first detector at 10 m from the pinch. Each signal is a PMT measurement that has been digitized and scaled to have peak value 1.

at the bottom of Fig. 7. There is a higher density of neutrons below the 2 MeV level (shown in bright yellow) for the reconstruction using the uncorrected ToF data and consequently, the density of neutrons near 2.45 MeV is reduced. Essentially, leaving the scatter field uncorrected downward biases the

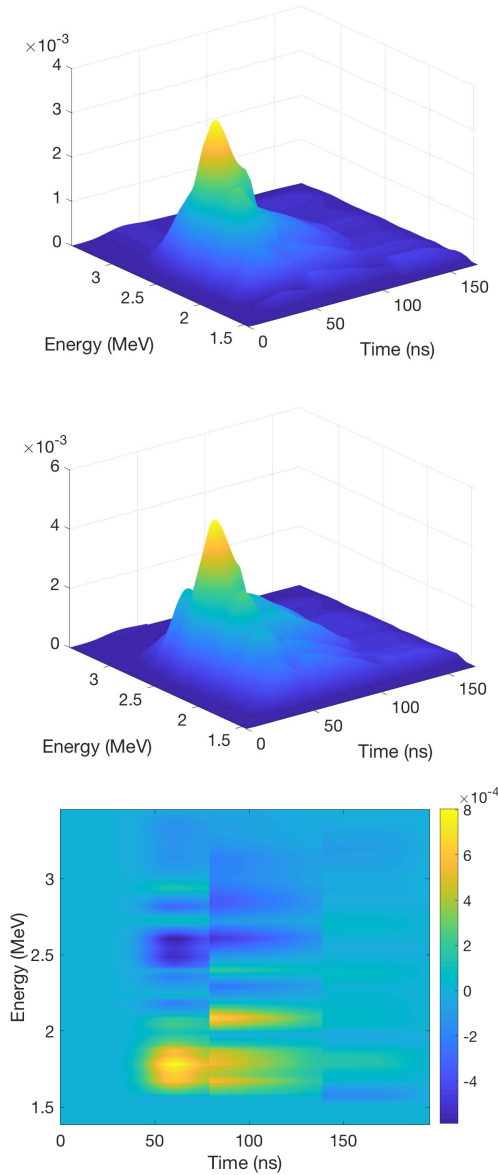


Fig. 7. Reconstruction of Trial 1 using the detector data that has been corrected by subtracting the background measurement (top) compared the reconstruction using the data prior to correcting for the scatter field (middle) and the difference between the two reconstructions (bottom). The discontinuities across the time axis correspond to the endpoints of the three subintervals, τ_k .

energy density, with a maximum relative difference of 23%. Correcting for the scatter field as we have done using the shadow bar measurement systems mitigates this bias in the reconstruction.

C. Comparison of Maximum Neutron Energy Calculations

For a simple validation of this approach to neutron time-energy spectrum reconstruction, we compare the maximum neutron energy for a given shot, computed from the densities shown in Fig. 4, to the energy associated with the separation between gamma and neutron arrival at a single detector. To measure the separation between the two pulses, we find the times of the 10% rise of the leading edge of the gamma and neutron pulses; see Fig. 8. The time separation

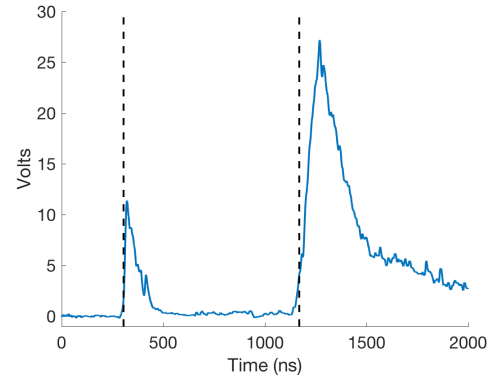


Fig. 8. Example neutron/gamma pulse measured with an FG detector at 22 m from the pinch. The black dashed lines indicate the estimated arrival of the gamma and neutron pulses.

TABLE IV
PREDICTED MAXIMUM ENERGY ACCORDING TO (9) AND THE
TOMOGRAPHICALLY RECONSTRUCTED MAXIMUM
ENERGY FOR EACH OF THE FOUR TRIALS

Trial	Predicted Max Energy (MeV)	Reconstructed Max Energy (MeV)
1	2.90	2.89
2	3.00	3.04
3	3.01	3.04
4	3.00	3.01

between the pulses gives the velocity of the neutrons at a detector d meters from the pinch

$$v = \frac{d}{t_\gamma + \Delta t} \quad (9)$$

where t_γ is the time for the gamma rays to travel d meters ($t_\gamma = d/c$, where c is the speed of light), Δt is the time separation between the gamma and neutron pulses, and kinetic energy is given by $E = (1/2) mv^2$.²

It is important to recognize that using the leading edge of the pulses to measure the separation between the gamma and neutrons biases the energy estimation, that is, we are obtaining an energy estimation of the highest energy neutrons. As the neutron pulse travels over distance, the higher energy neutrons overtake the lower energy neutrons, thus the leading edge of the pulse will be made up of an increasing proportion of high energy neutrons. Therefore, we typically estimate higher energies at greater distances for a given pinch. The estimated maximum neutron energy found according to (9) and the maximum energy in the reconstructed spectrum are compared in Table IV. In each case, the two methods for obtaining the maximum energy agree well.

V. CONCLUSION

Reconstructing the time-energy density of the fusion pinch is notoriously difficult due to large scatter fields in many experimental facilities and the blurring of the neutron pulse caused by ToF spreading. We have developed a novel approach

²This is appropriate, since 2.45-MeV neutrons are essentially nonrelativistic.

to neutron tomography for a DPF that ameliorates both of these issues. First, the use of shadow bar detector system pairs significantly improves the quality of the spectrum reconstructions, as it removes the downward bias in neutron energy introduced into the measured signals by scattered neutrons. Second, incorporating a direct measurement of the neutron pulse time profile near the origin of the pinch provides a constraint that allows to reconstruct the time-dependent neutron energy spectrum at time resolution commensurate with the resolution of the measurement system and energy resolution estimated to be finer than 100 keV, using data collected by neutron ToF detectors at only four distinct distances. This also allows to reconstruct multiple pinches with different energy profiles, which is not possible for previous work based on fully separable tomographic models. These two improvements provide a robust new approach to neutron ToF spectroscopy.

APPENDIX DETECTOR SPECIFICATIONS

The ToF detectors in each shadow bar pair are composed of a plastic scintillator coupled with a PMT used to convert particle flux into a recordable current source. The scintillator is a 2-in. long, 1-in. diameter cylinder of EJ-232 plastic scintillator which emits UV light at 370 nm, and has a 350 ps rise time with a 1.6 ns decay time. The PMT is a Hamamatsu R5924, 19-stage mesh-type PMT, with 2.5 ns anode rise time, 9.5 ns transit time through the PMT, and 440 ps transit time spread. When the neutrons pass through the scintillator, the organic solvents in the plastic absorb some of the particle's energy and re-emit that energy as optical photons. The photons then hit the photocathode of the PMT and knock electrons loose. Each stage of the PMT is biased with high voltage, and, when the electrons are freed from the cathode, the electric field created by the stage biasing carries the electrons toward the first dynode. When the electrons strike the dynode, additional electrons are produced and captured by the electric field, continuing through 19 stages and giving the PMT an overall gain on the order of 1×10^7 . In order to control the bias voltage on each stage and supply the charge to the dynodes, custom printed circuit boards (PCBs) are used along with Stanford Research Systems high voltage power supplies.

The detector used for the close-range measurement is a Hamamatsu R1193U ultrafast photodetector, which has rise- and fall-times of 270 and 100 ps, respectively. This photodetector platform has a voltage-independent gain of unity and a maximum radiant sensitivity of 2×10^{-2} mAW. Compared to PMTs, with gains on the order of 10^7 and radiant sensitivity of 10^2 mAW, the photodetector is a relatively low-sensitivity diagnostic. To compensate for this, the photodetector was positioned as close as practical to the exterior surface of the vacuum chamber of the DPF system, maximizing the neutron flux to the detector.

The photodetector in the close-range system was housed in an enclosure of 0.5 in. of stainless steel and 0.25 in. of lead. An Eljen EJ-232 plastic scintillator measuring 1-in. diameter, 1-in. long was coupled to the photodetector to convert the neutron pulse to primarily ultraviolet light. A supply voltage

of +1200 V was used to power the photodetector and the high voltage and signal cables were both RG-223 coaxial transmission lines approximately 25-ft in length. A 1-GHz LeCroy HDO 6104 digitizer with 2.5 GS/s sampling rate and a nominal 12-bits of quantization was used to capture the waveform of the detector. To shield against RF interference, the digitizer was housed in an electromagnetic interference (EMI) screen box and the cables were enclosed in Glenair conduit, connecting the screen box to the stainless steel detector enclosure.

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REFERENCES

- [1] N. Filipov, T. Filipova, and V. Vinogradov, "Dense, high-temperature plasma in a noncylindrical 2-pinch compression," *Nucl. Fusion*, vol. 2, Suppl. 577, 1962. [Online]. Available: https://www-pub.iaea.org/MTCD/Publications/PDF/TE_1708_web.pdf
- [2] J. W. Mather, "Formation of a high-density deuterium plasma focus," *Phys. Fluids*, vol. 8, no. 2, p. 366, 1965.
- [3] B. Appelbe and J. Chittenden, "Understanding neutron production in the deuterium dense plasma focus," *AIP Conf.*, vol. 1639, no. 1, pp. 9–14, 2014.
- [4] B. Appelbe and J. Chittenden, "Neutron spectra from beam-target reactions in dense Z-pinch," *Phys. Plasmas*, vol. 22, no. 10, Oct. 2015, Art. no. 102703.
- [5] B. T. Meehan and J. H. J. Niederhaus, "Fully three-dimensional simulation and modeling of a dense plasma focus," *J. Defense Model. Simul.*, vol. 13, no. 2, pp. 153–160, Apr. 2016.
- [6] A. Schmidt, V. Tang, and D. Welch, "Fully kinetic simulations of dense plasma Focus-Z-pinch devices," *Phys. Rev. Lett.*, vol. 109, no. 20, Nov. 2012, Art. no. 205003.
- [7] A. Schmidt, A. Link, D. Welch, J. Ellsworth, S. Falabella, and V. Tang, "Comparisons of dense-plasma-focus kinetic simulations with experimental measurements," *Phys. Rev. E, Stat. Phys. Plasmas Fluids Relat. Interdiscip. Top.*, vol. 89, no. 6, Jun. 2014, Art. no. 061101.
- [8] H. Brysk, "Fusion neutron energies and spectra," *Plasma Phys.*, vol. 15, no. 7, pp. 611–617, Jul. 1973.
- [9] I. Tiseanu and T. Craciunescu, "Evaluation of reconstruction methods for time-resolved spectroscopy of short-pulsed neutron sources," *Nucl. Sci. Eng.*, vol. 122, no. 3, pp. 384–394, Mar. 1996.
- [10] M. Vlad, "A time resolving spectrometry method for particles emitted in intense bursts," *Nucl. Instrum. Methods Phys. Res. A, Accel. Spectrom. Detect. Assoc. Equip.*, vol. 227, no. 2, pp. 327–334, Nov. 1984.

- [11] M. Vlad, M. Ganciu, M. Ivaşcu, N. Mandache, F. Spineanu, and V. Zoiţa, "Preliminary results on time resolving spectrometry for neutrons emitted in plasma focus discharges," *Nucl. Instrum. Methods Phys. Res. A, Accel. Spectrom. Detect. Assoc. Equip.*, vol. 274, no. 3, pp. 613–615, Jan. 1989.
- [12] A. Roomi, M. Habibi, E. Saion, and R. Amrollahi, "Modified reconstruction of neutron spectrum emitted in dense plasma focus devices by MCNP code and Monte-Carlo method," *J. Fusion Energy*, vol. 30, no. 1, pp. 78–82, Feb. 2011.
- [13] T. Tiseanu, G. Decker, and W. Kies, "A Monte-Carlo technique for the reconstruction of time dependent spectra of short-pulse neutron sources," *Nucl. Instrum. Methods Phys. Res. A, Accel. Spectrom. Detect. Assoc. Equip.*, vol. 373, no. 1, pp. 73–80, Apr. 1996.
- [14] L. Soto, "New trends and future perspectives on plasma focus research," *Plasma Phys. Controlled Fusion*, vol. 47, no. 5A, pp. A361–A381, May 2005.
- [15] N. Bennett *et al.*, "Kinetic simulations of gas breakdown in the dense plasma focus," *Phys. Plasmas*, vol. 24, no. 6, Jun. 2017, Art. no. 062705.



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